

Paper 2 - Set 09 - Solutions

Generated practice paper: answer key and worked solutions.

Answer key

Question	Answer	Topic
1	E	Indices and surds
2	B	Binomial expansion
3	C	Simultaneous equations and inequalities
4	A	Identifying errors in proofs
5	E	Chains of reasoning
6	A	Identifying errors in proofs
7	G	Ratio, proportion and percentages
8	D	The logic of arguments
9	D	Mathematical proof
10	B	The logic of arguments
11	D	Pythagoras, mensuration and 3D
12	B	Mathematical proof
13	H	Polynomials, factor and remainder theorems
14	E	Chains of reasoning
15	B	Simultaneous equations and inequalities
16	E	Chains of reasoning
17	G	Graphs of functions
18	H	Statistics
19	C	Probability
20	H	The logic of arguments

Worked solutions

Question 1 IND-05 · Indices and surds · spec MM1.2

Simplify $\sqrt{75} - \sqrt{27} + \sqrt{12}$.

Answer: E

$$\sqrt{75} = 5\sqrt{3}, \sqrt{27} = 3\sqrt{3}, \sqrt{12} = 2\sqrt{3}, \text{ so the total is } (5 - 3 + 2)\sqrt{3} = 4\sqrt{3}.$$

Question 2 BIN-10 · Binomial expansion · spec MM2.4

Find the coefficient of x^5 in the expansion of $(1+x)^3(1+x)^4$.

Answer: B

$$(1+x)^3(1+x)^4 = (1+x)^7, \text{ and } \binom{7}{5} = 21.$$

Question 3 SIM-09 · Simultaneous equations and inequalities · spec MM1.4

The simultaneous equations $y = x^2 + 2x + c$ and $y = 4x + 1$ have exactly one solution. Find the value of c .

Answer: C

$$x^2 - 2x + (c - 1) = 0 \text{ has one solution when } 4 - 4(c - 1) = 0, \text{ so } c = 2.$$

Question 4 ERR-09 · Identifying errors in proofs · spec Err2

A student proves that the equation $x^2 + 2x + 3 = 0$ has no real solutions.

I Complete the square: $x^2 + 2x + 3 = (x + 1)^2 + 2$.

II $(x + 1)^2 \geq 0$ for all real x .

III So $(x + 1)^2 + 2 \geq 2 > 0$ for all real x .

IV Hence the equation has no real solutions.

Which of the following is correct?

Answer: A

Every step is valid.

Question 5 RSN-21 · Chains of reasoning · spec Prf5

Seven consecutive integers are written down. Which of the following statements is/are true?

I Their sum is divisible by 7.

II Their product is divisible by 7.

III Their sum is odd.

Answer: E

The sum is $7m$ where m is the middle integer (I). Seven consecutive integers cover every remainder mod 7, so one is a multiple of 7 (II). $7m$ is odd only when m is odd, so III need not hold.

Question 6 ERR-12 · Identifying errors in proofs · spec Err2

A student solves $x^2 = 4x$.

I Divide both sides by x : $x = 4$.

II So the only solution is $x = 4$.

Which of the following is correct?

Answer: A

Factorise instead: $x(x - 4) = 0$ gives $x = 0$ or $x = 4$.

Question 7 RAT-15 · Ratio, proportion and percentages · spec M3.8

Which of the following statements is/are true?

I Increasing a quantity by 50% and then decreasing the result by 50% returns it to its original value.

II A 10% increase followed by a 10% increase is a 21% increase overall.

III If A is 25% more than B , then B is 20% less than A .

Answer: G

I is false: $1.5 \times 0.5 = 0.75$. II: $1.1^2 = 1.21$, true. III: $A = 1.25B \Rightarrow B = 0.8A$, true.

Question 8 LGC-03 · The logic of arguments · spec Arg4

Which of the following is the negation of the statement '**every student in the class scored more than 50'?

Answer: D

The negation of 'for all x , $P(x)$ ' is 'there exists x such that not $P(x)$ '. 'Not more than 50' means '50 or less'.

Question 9 PRF-07 · Mathematical proof · spec Prf1

Prove by cases that $n^3 - n$ is divisible by 3 for every integer n . Which set of cases is both sufficient and necessary to consider?

Answer: D

$n^3 - n = (n - 1)n(n + 1)$ is a product of three consecutive integers; splitting by remainder on division by 3 shows one factor is a multiple of 3 in every case.

Question 10 LGC-22 · The logic of arguments · spec Arg2

For real numbers a, b : P : ' $ab = 0$ '; Q : ' $a = 0$ '. Which one of the following is true?

Answer: B

$a = 0 \Rightarrow ab = 0$, so P is necessary for Q . But $ab = 0$ could be because $b = 0$ with $a \neq 0$, so P is not sufficient.

Question 11 X2-29 · Pythagoras, mensuration and 3D · spec M5.16

P: 'a solid cylinder and a solid cube have equal volumes'. *Q*: 'the cylinder and the cube have equal surface areas'. Which one of the following is true?

Answer: D

Equal volumes say nothing about surface areas (a long thin cylinder has a large surface area), and equal surface areas say nothing about volumes. Neither implies the other.

Question 12 PRF-13 · Mathematical proof · spec Prf1

Which of the following is a correct proof that **the sum of any three consecutive integers is divisible by 3**?

Answer: B

A is general and correct. B checks cases; C uses non-consecutive integers; D's reasoning is invalid (containing a multiple of 3 does not make the sum a multiple of 3).

Question 13 POL-08 · Polynomials, factor and remainder theorems · spec MM1.6

Which of the following statements about the polynomial $p(x) = x^3 - 6x^2 + 11x - 6$ is/are true?

I $p(x)$ has three distinct real roots.

II $(x - 4)$ is a factor of $p(x) - 6$.

III $p(x) > 0$ for all $x > 3$.

Answer: H

$p(x) = (x - 1)(x - 2)(x - 3)$, so I is true. $p(4) - 6 = 3 \cdot 2 \cdot 1 - 6 = 0$, so II is true. For $x > 3$ all three factors are positive, so III is true.

Question 14 RSN-04 · Chains of reasoning · spec Prf5

x and y are positive integers with $x^2 - y^2 = 45$. How many possible pairs (x, y) are there?

Answer: E

$(x - y)(x + y) = 45$ with $0 < x - y < x + y$: (1, 45), (3, 15), (5, 9) give $(x, y) = (23, 22), (9, 6), (7, 2)$. Three pairs.

Question 15 SIM-18 · Simultaneous equations and inequalities · spec MM1.5

Consider the statement: for all real x , if $x > 2$ then $x^2 - 3x + 2 > 0$. Which of the following is true?

Answer: B

$x^2 - 3x + 2 = (x - 1)(x - 2) > 0$ exactly when $x < 1$ or $x > 2$. So $x > 2$ does imply it (true), but the converse fails, e.g. $x = 0$ gives $2 > 0$ without $x > 2$.

Question 16 X2-26 · Chains of reasoning · spec Prf5

The positive integers a , b and c satisfy $a < b < c$ and $a + b + c = 12$. Which of the following statements must be true?

- I $a \leq 3$
- II $c \geq 5$
- III $b = 4$

Answer: E

Since $a < b < c$, $3a < 12$ so $a \leq 3$ (I); $3c > 12$ so $c \geq 5$ (II). III fails: $1 + 2 + 9 = 12$ has $b = 2$.

Question 17 X2-21 · Graphs of functions · spec MM8.6

$p(x)$ is a polynomial of degree 4 with real coefficients. Which of the following statements must be true?

- I $p(x) = 0$ has at least one real root.
- II $p(x) = 0$ has at most four real roots.
- III The graph of $y = p(x)$ has at least one turning point.

Answer: G

I fails: $x^4 + 1$ has no real roots. II: a degree-4 polynomial has at most 4 roots. III: an even-degree polynomial tends to the same infinity at both ends, so it must turn at least once.

Question 18 X2-28 · Statistics · spec M6.3

A set of five distinct positive integers has median 10 and mean 10. Which of the following statements must be true?

- I The largest value is at least 12.
- II The smallest value is at most 8.
- III The range is at least 4.

Answer: H

The two values above the median are distinct integers greater than 10, so at least 11 and 12 (I); similarly the two below are at most 9 and 8 (II); hence the range is at least $12 - 8 = 4$ (III).

Question 19 X2-19 · Probability · spec M7.7

Events A and B have $P(A) > 0$ and $P(B) > 0$. P : ' A and B are mutually exclusive'. Q : ' A and B are not independent'. Which one of the following is true?

Answer: C

Mutually exclusive gives $P(A \text{ and } B) = 0 \neq P(A)P(B)$, so they cannot be independent: P is sufficient. But two overlapping events can also fail to be independent, so P is not necessary.

Question 20 LGC-09 · The logic of arguments · spec Arg1

Consider the statement S : **for all real x , if $x > 0$ then $x + \frac{1}{x} \geq 2$.** Which of the following statements is/are true?

- I** S is true.
- II** The converse of S (for all real x , if $x + \frac{1}{x} \geq 2$ then $x > 0$) is true.
- III** The contrapositive of S is true.

Answer: H

For $x > 0$, $x + \frac{1}{x} - 2 = \frac{(x-1)^2}{x} \geq 0$, so S is true, and so is its contrapositive (III). For the converse: if $x < 0$ then $x + \frac{1}{x} < 0 < 2$, so $x + \frac{1}{x} \geq 2$ forces $x > 0$ (II true).